

Malavika Ajith

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Summary

B.Tech Computer Science Engineering student with a strong foundation in problem solving, data structures, algorithms, and object-oriented programming. Skilled in Python, C++, Java, and SQL, with hands-on experience building scalable applications and AI/ML-driven solutions through academic projects and internships. Familiar with RDBMS (PostgreSQL, MySQL), NoSQL concepts, and modern web development. Adept at debugging, performance optimization, and technical troubleshooting in fast-paced, collaborative environments. Passionate about delivering robust, efficient, and maintainable software solutions.

Education

Amrita Vishwa Vidyapeetham, Bachelor of Technology · Computer Science Engineering · Kollam	Nov 2022 - Present 7.98/10
Christ Central School 12th Grade · CBSE · Thiruvalla	Jun 2021 - Mar 2022 85.2 %
Chinmaya Vidyalaya 10thGrade · CBSE · Chengannur	Jun 2019 - Mar 2020 95.4%

Experience

Infospica Web Developer · Web Development Intern · Internship	Aug 2024 - Aug 2024 Trivandrum
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Collaborated with the development team to design, build, and optimize web applications focused on user experience, performance, and security. Gained practical experience in front-end development using WordPress, including customization and plugin integration. Learned to implement secure authentication and safeguard user data, strengthening site security protocols.

Hospital Booking Website (WordPress) | Link

- Developed a user-friendly website for Asclepius Hospital to enable efficient appointment booking and easy access to medical services and hospital information.
- Designed and implemented a secure patient portal for accessing medical records and consultation history, ensuring data privacy and compliance with healthcare standards.
- Delivered a platform reflecting the hospital's dedication to quality healthcare through intuitive, accessible UI/UX, robust backend functionality, and code developed to rigorous coding standards.

Projects

AutoPulse-Intelligent Vehicle Lifecycle & Anomaly Management System	July 2025- present
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- Java, Spring Boot, React.js, PostgreSQL, Flask, Isolation Forest, Docker
- Developed a full-stack web application to manage vehicle data and lifecycle tracking with real-time anomaly detection powered by Python-based ML models.
 - Build secure and scalable backend services in Java using Spring Boot, integrating PostgreSQL for persistent storage and role-based access control.
 - Implementing anomaly detection with Isolation Forest via a Flask microservice, triggering "Anomaly Detected" alerts on a live React.js dashboard.
 - Containerize frontend, backend, and ML services using Docker Compose, enabling smooth microservice and environmental consistency.

Deepfake Detection System	Apr 2025 - June 2025
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- PyTorch, Grad-CAM, SHAP, OpenCV, ViT, FaceForensics++, ONNX
- Developed algorithms to detect manipulated video and image content by analyzing subtle visual artifacts and inconsistencies.
 - Utilized state-of-the-art deep learning models (e.g., Vision Transformers) and computer vision techniques to distinguish deepfakes from authentic media.
 - Strengthened multimedia misinformation defense by integrating detection capabilities into broader content verification workflows.

Bias-Aware & Explainable Fake News Detection System	Apr 2025 - June 2025
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- RoBERTa , SHAP, LIME Role: Business Analyst (within Scrum Team)
- Designed and implemented an AI model leveraging deep learning architecture (RoBERTa) for high-accuracy fake news classification.
 - Applied bias mitigation techniques to ensure fairness and reduce model prejudice.
 - Integrated explainability tools like SHAP and LIME to make model decisions transparent and trustworthy.
 - Created comprehensive documentation, including system architecture, algorithm specifications, and procedures for installation and maintenance, facilitating seamless deployment and team collaboration

- Developed SRL-TS, a robust framework for secure time synchronization in distributed systems, integrating Reinforcement Learning (PPO) and autoencoder-based anomaly detection to dynamically adjust sync intervals and detect malicious time updates.
- Implemented multi-node timestamp communication and control using TCP socket programming, enhancing resilience against adversarial attacks like spoofing and Byzantine faults.
- Designed a Trust Score System to evaluate node reliability, effectively isolating malicious nodes and maintaining high synchronization accuracy (achieved <2ms offset in normal conditions, >92% anomaly detection accuracy).

Certifications

Cyber Security and Ethical Hacking Coincent and Microsoft Took part in a Course in Cyber Security and Ethical Hacking at Coincent (2023)	Mar 2023 -
Evolumin Hackathon ACM Student Chapter Amritapuri National-Level Hackathon (Amrita Vishwa Vidyapeetham, 2024)	Nov 2024 -
CompTIA Network+: Network Services & Data Center Architectures Infosys Springboard Covered network services, data center architectures, and enterprise infrastructure Gained foundational knowledge in network security principles	Mar 2025 -
Problem Solving (Intermediate) HackerRank Demonstrated proficiency in data structures and algorithmic thinking.	Apr 2025 -

Social Outreach Programme

Student Social Responsibility (SSR)
Conducted a workshop on Web Development and Career Guidance for Grade 11 students at Lightland Higher Secondary School, Oachira. Conducted a hands-on workshop covering HTML, CSS, and career guidance for higher secondary students.

Technical & Interpersonal Skills

Programming & Scripting
Python, C, C++, Java(basics),Bash Scripting(with experience in Unix/Linux environments)

Database
PostgreSQL,MySQL

Web & Systems Development
HTML, CSS, JavaScript, WordPress

Cybersecurity & Ethical Hacking
Encryption(OpenSSL),Basic Vulnerability Analysis, Secure Authentication

Cloud Platforms :
AWS (currently enrolled in the Academic Foundation Course)

Tools:
OpenSSL,npm, Docker

Leadership & Involvement
Leadership,Team Management, Event Coordination

Languages
English , Malayalam, Hindi: Fluent to Read, Write, and Speak